Inference-Time Solver-Verifier Debate Exceeds Pass@k on Math Reasoning

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Abstract

Recent work argues that multi-agent debate at inference time fails to beat self-consistency once the two are compared at matched generation budget. We revisit this on Qwen-family math reasoning with a four-turn Solver-Verifier protocol in which one model plays both roles, differentiated only by system prompt. On three checkpoints (Qwen3-1.7B base, Qwen3-4B base, and Qwen3-4B after reinforcement-learning-with-verifiable-rewards (RLVR) training) the protocol beats $k = 4$ self-consistency on MATH-500 by 21.2, 22.4, and 9.8 percentage points, with non-overlapping paired bootstrap CIs. More importantly, it exceeds the pass@k oracle, the theoretical ceiling for any aggregator over k parallel samples, by five to thirty points depending on problem difficulty, with the largest single-cell margin on base-model level-5 MATH (0.769 vs 0.470). The Verifier turn must therefore be generating new trajectories via critique-conditioning rather than selecting among existing ones. A frozen base model in the Verifier role produces a statistically indistinguishable lift, indicating the scaffold structure carries the value rather than any role-specialized training. The same scaffold also recovers AIME 2025 base-model accuracy from 2/30 single-shot and 0/30 self-consistency (where majority vote concentrates on the dominant wrong mode) up to 10/30. As a complementary control we train a multi-agent reward design with class-reweighted verdict judgment and a counterfactual Verifier-value channel; the resulting Solver is indistinguishable from a single-agent GRPO baseline at matched compute. For this benchmark and family, the productive multi-agent work happens at inference time, not in the training objective.

1 Introduction

The literature on multi-agent debate for language-model reasoning has split into two camps. The first treats multiple cooperating or adversarial LLM agents as a productive inference-time scaffold and reports gains over single-agent baselines on a range of reasoning tasks [1–6]. The second argues that those gains evaporate, or invert, once one controls for compute: matched-token generation budgets, longer chain-of-thought, and especially self-consistency at k samples have been shown to close the gap on several setups [7–9]. The critique literature has largely settled on self-consistency at matched compute as the load-bearing single-agent baseline that any multi-agent claim must clear.

We test this empirically on a specific scaffold and a specific model family, and find that the critique conclusion does not generalize. Our scaffold is a strict four-turn Solver-Verifier-Solver-Verifier debate with role differentiation supplied entirely by system prompts; no architectural asymmetry, no Verifier training. We pre-register a falsification gate before any data is seen: for the scaffold to count as a positive finding, it must beat both single-shot and $k = 4$ self-consistency by at least three percentage points with non-overlapping 95% bootstrap CIs on at least one of MATH-500 L4, L5, AIME 2024, AIME 2025, or GPQA-Diamond.

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The gate clears on all three Qwen-family checkpoints we test. On the untrained Qwen3-4B base the scaffold beats matched-compute self-consistency by 22.4 points overall, 28 points on level 4, and 41 points on level 5; on a single-agent GRPO-trained checkpoint the lift is smaller (9.8 points overall) because the matched-compute baseline is already stronger, but the margin remains gate-clearing. Both checkpoints converge to roughly 0.83 on MATH-500 under the scaffold despite a 12-point single-shot gap, which says something specific about what the scaffold is doing: RLVR closes part of the gap to debate, but only part of it.

We make four contributions. First, we release a five-arm matched-compute evaluation matrix (Section 3) covering single-shot, two debate variants (one checkpoint serving both roles; an asymmetric variant with a frozen base Verifier), self-consistency at $k = 4$, and the pass@k oracle. The code is open-source and runs against any HuggingFace-format checkpoint. Second, we show that the debate gain exceeds the pass@k oracle, the theoretical ceiling for any aggregator over k parallel samples (Section 5.2). The Verifier turn must therefore be generating new trajectories rather than selecting among existing ones. Third, we characterize when self-consistency works and when it fails: it gives a consistent eight-point lift on MATH-500 across all our checkpoints but drops base-model AIME 2025 accuracy from two-of-thirty to zero-of-thirty when the model's modal samples are wrong (Section 5.5). Debate is robust to this failure mode. Fourth, we run an ambitious training-time multi-agent design as a complementary negative (Section 6): with role-asymmetric multi-channel rewards including a class-reweighted verdict signal [10] and a counterfactual Verifier-value channel [11], the resulting Solver is indistinguishable from one trained by ordinary single-agent GRPO at matched compute. The productive multi-agent work, for this benchmark and family, happens at inference time.

2 Related Work

Multi-agent debate as inference scaffolding. [1–6, 10, 12] train or compose multiple LLM agents. The methodological critique line [8, 7, 9] argues these gains disappear under matched-compute controls. Our result on Qwen3-4B contradicts this critique on a specific slice their experimental grids do not cover. The closest published cell to ours is (Llama3.1-8B, full MATH n=5000, $k \approx 6$ self-consistency, symmetric MAD methods) in [7]: in that cell, self-consistency beats every MAD method by 2-32pp. Our cell (Qwen3-4B, MATH-500 + AIME, $k = 4$ self-consistency, role-asymmetric Solver-Verifier debate) differs on three jointly-unfilled axes: model family, protocol structure, and difficulty stratification. Each is a plausibly load-bearing difference: Qwen-family models are unusually RLVR- and prompt-sensitive [13]; role-asymmetric verdict-emitting protocols have no precedent in the MAD-critique literature; and MATH-500 levels 4-5 (where our largest gains sit) are systematically harder than the average of MATH-5000.

Concretely, the critique papers test the following non-overlapping setups:

- [7]: 5 symmetric MAD methods (SoM, Multi-Persona, Exchange-of-Thoughts, ChatEval, AgentVerse) $\times$ 4 models (GPT-4o-mini, Claude-3.5-Haiku, Llama3.1-8B, Llama3.1-70B) $\times$ 9 benchmarks (MMLU, MMLU-Pro, CommonsenseQA, ARC-Challenge, AGIEval, GSM8K, MATH-5000, HumanEval, MBPP). No Qwen, no MATH-500 stratification, no AIME, no role-asymmetric debate, no $k = 4$ token-matched SC.
- [9]: 5 MAS architectures $\times$ 3 large models (Qwen3-30B-A3B, DeepSeek-R1-Distill-Llama-70B, Gemini-2.5) $\times$ 2 multi-hop QA benchmarks (FRAMES, MuSiQue-4hop). No 4B model, no math.
- [8]: AFlow/OneFlow workflow methods on Qwen-3 8B / GPT-4o-mini. "Strong single-agent baseline" is "same workflow, single executor with KV reuse", not self-consistency at $k = 4$. MATH: OneFlow 54.1 vs AFlow 55.6 ($\approx$ null).

We do not claim our finding overturns the critique literature's general conclusion. We claim it identifies a specific scope-restricted regime (small Qwen-family models, role-asymmetric protocol, hard-math difficulty band) where the conclusion does not hold. This is consistent with, not contradictory to, the broader observation that *most* multi-agent debate configurations fail their matched-compute tests.

Multi-agent debate distillation. IMAD [14], AgentArk [15], and Chain-of-Agents [16] distill debate trajectories into a single model via post-debate SFT or RL. Our setting is orthogonal: we measure the inference-time-only marginal value of debate on policies that have *not* been distilled. Our findings suggest the distillation effort may be lower-value than the literature assumes, since the un-distilled base already supports a strong debate scaffold at inference.

Training-time multi-agent RL. MARTI, Stronger-MAS, AT-GRPO, MAPoRL, CRM, MAGRPO [1, 2, 4, 3, 5, 17] train multiple agents with joint or per-agent rewards. None separate the Verifier's judgment quality from its answer (a single-channel reward conflates the two), and none use per-channel normalization à la GDPO [18]. DTCA (§3.2) is the strongest single-paper implementation of role-asymmetric multi-channel rewards in this lineage, and produces a Solver indistinguishable from single-agent GRPO at matched compute. This complements our positive inference-time finding: the value of multi-agent in this regime is *not* in the training-time reward design.

Self-consistency and parallel sampling. Self-consistency [19] samples k responses and majority-votes the answer; pass@k is the oracle aggregator. Both are strong matched-compute baselines and the critique literature's primary single-agent stand-in. We use both, and we report a failure mode of self-consistency that the critique literature does not address. On uniformly-hard problems (our AIME 2025 base result: 6.7% single-shot, 0.0% at $k = 4$), self-consistency catastrophically fails, because majority voting concentrates on the dominant wrong mode. Debate lifts the same cells (base: 0.0% to 23.3%; solo-GRPO: 13.3% to 36.7%).

RLVR limits and base-model analyses. [13] argues RLVR improves pass@1 but does not expand the underlying reasoning capacity beyond the base. Our results are consistent with this: the base model's *latent* debate capacity is roughly equivalent to the RLVR-trained model's debate capacity (both hit $\sim$0.83 on MATH-500 with the same debate scaffold), even though their single-shot performance differs by 12pp. RLVR concentrates probability mass on outcome-correct trajectories; this raises single-shot accuracy but does not change the upper bound that a critic-scaffold can extract.

3 Method

3.1 Trained Solvers

We compare two trained policies on the same architecture (Qwen3-4B):

- **Base.** The untrained `Qwen/Qwen3-4B` instruct checkpoint.
- **Solo-GRPO.** Same architecture, trained with outcome-only single-agent GRPO on MATH-500 levels 3-5 for 30 PPO updates (full config in Appendix A): $K = 8$ rollouts/problem, response budget 2048, KL-anchored at $\lambda_{KL} = 0.001$ with `low_var_kl` [20], repetition penalty 1.15. Trained on Modal 2× H200, $\sim$7 hours wall-clock.

3.2 DTCA recipe (complementary negative result)

We additionally train Qwen3-4B with DTCA, our role-asymmetric multi-channel reward design, as a controlled comparison for the multi-agent training-time literature:

- **Protocol v2.** A two-agent debate (Solver, Verifier) with 4 turns total. Each Verifier turn emits *both* a `<verify>approve|reject</verify>` verdict and a `\boxed{}` answer. The verdict is rewarded for judgment quality (R_verdict); the Verifier's `\boxed{}` answer is *not directly rewarded*, only via the shared-trajectory channels.
- **R_verdict with class reweighting** (Self-Verify Eq. 5 [10]) for the imbalanced-base-rate case.
- **R_verifier_value.** Counterfactual contribution channel à la Dr. MAMR [11], using Solver turn-0 as a cheap counterfactual proxy.
- **GDPO per-channel advantage normalization** [18].
- **Raw-bypass format penalty.** Verifier turns missing `<verify>` or `\boxed{}` receive a per-turn -1.0 raw penalty that bypasses GDPO normalization. Necessary because under format collapse, the channel's empirical variance approaches zero and the in-channel penalty becomes toothless. A transferable design pattern for normalized multi-channel RL with mixed-magnitude reward channels.

3.3 Five-arm inference matrix

Given a trained Solver checkpoint θ, we evaluate five inference arms:

- **A. single-shot.** θ produces one solution with the Solver system prompt. $k = 1$.
- **B. debate-self.** θ plays both roles in a 4-turn Solver-Verifier-Solver-Verifier debate. Role differentiation: system prompts only. Solver: "Solve the problem step by step. End with `\boxed{answer}`." Verifier: "Critique the Solver's solution. Output `<verify>approve</verify>` or `<verify>reject</verify>` then `\boxed{your answer}`." $k = 1$.
- **C. debate-asymmetric.** Solver $= \theta$, Verifier $= \theta_{\text{base}}$ (frozen Qwen3-4B base). Same 4-turn protocol. Tests whether the Verifier's RL training is necessary for the debate gain. $k = 1$. (Reported only for the solo-GRPO Solver in Section 5.4; the asymmetric arm with two co-located vLLM engines required the V0 engine to avoid the V1 KV-cache memory-accounting bug.)
- **D. self-consistency.** θ produces $k = 4$ samples; majority-vote on the normalized `\boxed{}` answer. Matched generation budget vs B ($4 \times 2048 = 8192$ tokens).
- **E. pass@k.** θ produces $k = 4$ samples; oracle any-correct aggregation. Matched budget upper bound.

All arms: per-role temperatures (Solver 0.7, Verifier 0.3), `repetition_penalty=1.15`, `max_tokens=2048`, sympy-aware answer matching. Full implementation: `dtca/eval/standalone.py` (open-source, 23 unit tests).

3.4 Falsification gate

Pre-registered: for inference-time debate to count as a positive finding, arm B or arm C must beat *both* A (single-shot) and D (self-consistency at $k = 4$) by $\geq$ 3pp with non-overlapping 95% bootstrap CIs on at least one of MATH-500 L4, L5, AIME24, AIME25, GPQA-Diamond. The matched-compute control (clause vs D) is the load-bearing constraint per [9].

4 Experimental Setup

Models. Three Qwen-family checkpoints for the inference-scaffold matrix: (i) Qwen3-1.7B base (cross-scale generalization), (ii) Qwen3-4B base (primary), (iii) Qwen3-4B + single-agent GRPO step-30 (this paper's "solo-GRPO"). One additional Qwen3-4B + DTCA step-30 checkpoint for the training-time multi-agent comparison in Section 6. Cross-family probes on microsoft/Phi-3.5-mini-instruct, HuggingFaceTB/SmolLM2-1.7B-Instruct, and mistralai/Mistral-7B-Instruct-v0.3 are reported as preliminary in Appendix E; each hit a different blocker (answer-format extraction, context-window overflow under 4-turn debate, and a vLLM tokenizer-compatibility init bug, respectively) and we report only the cells that ran cleanly.

Eval datasets. MATH-500 (full, $n = 500$, stratified L1–L5), AIME 2024 ($n = 30$), AIME 2025 ($n = 30$). GPQA-Diamond pending (gated HF dataset; auth setup in progress).

Eval pipeline. Standalone vLLM 0.8.4, V1 engine, bf16, on Modal 1× H200. Sympy-aware answer matching. Per-problem JSONL records preserved. Code: `dtca/eval/standalone.py`.

Bootstrap CIs. Per-cell 95% percentile bootstrap on the difference $p_B - p_D$, 2000 resamples per cell. For aggregate cells with $n = 500$ we cross-check against the standard Wald CI on the difference of proportions.

Reproducibility. Modal app IDs, exact configs, and reproduction scripts in Appendix B.

5 Results

5.1 Main result: debate beats matched-compute self-consistency on both base and RLVR-trained Solvers

Table 1 reports headline cells (MATH-500 overall, L4, L5, AIME 2024, AIME 2025) per checkpoint with all four inference arms side-by-side and a paired bootstrap CI on the $B - D$ margin. Full per-level accuracy and per-cell bootstrap CIs are in Appendix F.

Table 1: Headline cells. Arm A is single-shot, D is $k = 4$ self-consistency, E is the pass@4 oracle, B is 4-turn debate-self. $n = 500$ on MATH-500 overall; $n = 128$ on L4; $n = 134$ on L5; $n = 30$ per AIME year. Bootstrap CIs paired, 2000 resamples.

Cell	A	D	E	B	B−D	95% CI on B−D
Base Qwen3-4B						
MATH-500 overall	0.534	0.614	0.692	**0.838**	+22.4	[+18.6, +26.2]
MATH-500 L4	0.445	0.555	0.648	**0.836**	+28.1	[+20.3, +35.9]
MATH-500 L5	0.299	0.358	0.470	**0.769**	+41.0	[+32.1, +50.0]
AIME 2024	0.033	0.100	0.133	**0.367**	+26.7	[+13.3, +43.3]
AIME 2025	0.067	0.000	0.133	**0.333**	+33.3	[+16.7, +50.0]
Solo-GRPO step-30						
MATH-500 overall	0.656	0.734	0.780	**0.832**	+9.8	[+6.8, +13.0]
MATH-500 L4	0.617	0.695	0.773	**0.820**	+12.5	[+5.5, +20.3]
MATH-500 L5	0.500	0.567	0.597	**0.769**	+20.2	[+13.4, +27.6]
AIME 2024	0.133	0.100	0.167	**0.233**	+13.3	[+3.3, +26.7]
AIME 2025	0.100	0.133	0.133	**0.367**	+23.3	[+10.0, +40.0]
Base Qwen3-1.7B						
MATH-500 overall	0.520	0.564	0.674	**0.776**	+21.2	[+17.4, +25.2]
MATH-500 L4	0.477	0.484	0.633	**0.766**	+28.1	[+19.5, +35.9]
MATH-500 L5	0.261	0.299	0.433	**0.627**	+32.8	[+23.9, +41.8]
AIME 2024	0.000	0.033	0.033	**0.133**	+10.0	[0.0, +23.3]
AIME 2025	0.000	0.067	0.100	**0.267**	+20.0	[+6.7, +36.7]

The pre-registered gate (B beats D by at least three points with a non-overlapping CI) clears on every checkpoint and on multiple cells per checkpoint. The single largest margin is on base Qwen3-4B at MATH-500 L5: +41.0 points with CI [+32.1, +50.0]. The two base models produce statistically indistinguishable margins on MATH-500 overall (+21.2 on 1.7B vs +22.4 on 4B, overlapping bootstrap intervals) despite a 2.4× parameter gap; the debate-over-self-consistency margin is roughly invariant across that range.

Paired bootstrap is tighter than the Wald interval here because the within-problem correlation between arms B and D is positive: problems hard for one tend to be hard for both. On solo-GRPO L4, the Wald CI of [+2.1, +22.9] would have placed the lower bound below the three-point gate threshold; the paired bootstrap moves it to [+5.5, +20.3].

5.2 Debate exceeds the pass@k oracle

A more decisive test than self-consistency is the pass@k oracle: the fraction of problems where any of k independent samples is correct. By construction pass@k is the theoretical upper bound for any aggregation strategy over k parallel samples. If debate exceeds it, the scaffold is producing correct answers on problems where no independent sample is correct, ruling out the "more samples or better aggregation" explanation entirely.

The pass@4 oracle is reported as column E in Table 1. The $B - E$ margins, with 95% paired bootstrap CIs on the hardest cells, are summarized in Table 2; the full per-cell CI table is in Appendix F.

On every cell where the underlying model is below ceiling, debate exceeds the pass@k oracle. The aggregate gain is five points on the strongest single-shot checkpoint (solo-GRPO) and ten to fifteen points on the base models, growing with difficulty. On Qwen3-4B base MATH-500 L5, debate gets 30 percentage points more problems right than any of four independent samples does. On Qwen3-1.7B base AIME 2025, debate solves 27% of problems while pass@4 reaches only 10% and single-shot reaches zero.

Table 2: Hardest-cell $B - E$ summary. Full per-cell CIs in Appendix F.

Cell	$p_B - p_{\text{pass@4}}$	95% bootstrap CI
Base Qwen3-4B MATH-500 L5	**+29.9**	[+21.6, +38.1]
Base Qwen3-4B AIME 2025	+20.0	[+6.7, +36.7]
Solo-GRPO MATH-500 L5	+17.2	[+9.7, +24.6]
Solo-GRPO AIME 2025	+23.3	[+10.0, +40.0]
Qwen3-1.7B MATH-500 L5	+19.4	[+11.2, +26.9]
Qwen3-1.7B AIME 2025	+16.7	[+3.3, +30.0]

An aggregation explanation of debate's value cannot account for this. By construction no aggregator over the four parallel samples can do better than pass@4, and yet debate is doing exactly that. The Verifier turn changes the conditioning of the next Solver turn, producing trajectories that the model's temperature-0.7 sampling distribution does not contain. The strongest single-cell finding sits on base Qwen3-4B at MATH-500 L5, where debate's accuracy of 0.769 exceeds the pass@4 oracle ceiling of 0.470 by 29.9 points, with a bootstrap-CI lower bound of +21.6.

5.3 RLVR substitutes for inference-time debate; cross-scale invariance

The base Qwen3-4B model and the solo-GRPO-trained checkpoint converge to nearly identical accuracy under the 4-turn debate scaffold (0.838 vs 0.832 on MATH-500 overall). Their single-shot accuracies differ by 12.2 points and their self-consistency accuracies by 12.0 points, but those gaps collapse to under one point once debate is applied. On the hardest level, L5, both checkpoints reach exactly 0.769 in debate despite a 20-point gap in single-shot accuracy.

Adding the 1.7B base model into the picture extends the pattern across scale rather than across training. Under the same scaffold, Qwen3-1.7B base reaches 0.776 overall on MATH-500 with a B$-$D margin of +21.2 points (CI [+15.7, +26.7]). The 1.7B and 4B base margins are statistically indistinguishable on MATH-500 overall despite a 2.4× difference in parameters. The debate-versus-self-consistency margin does not scale with model size in this range.

A natural interpretation: the 4-turn scaffold extracts something close to an upper bound on the reasoning capacity available given the model's parametric knowledge. RLVR redistributes probability mass on individual samples but does not raise this ceiling. Smaller models hit lower ceilings overall (0.776 on 1.7B vs 0.838 on 4B base), but the margin that debate extracts over matched-compute self-consistency stays roughly constant. At this scale and benchmark, training-time outcome-only RLVR appears partially substitutable for inference-time debate effort: a practitioner without training compute can reach within one point of the RLVR-trained 4B checkpoint's debate accuracy by running the base 4B model in the same scaffold, or within six points using the base 1.7B model. Whether this picture extends to larger models, where debate has less headroom to fill, is an open question; the cross-checkpoint consistency at 1.7B–4B is the strongest evidence the current debate literature offers in that direction.

5.4 Asymmetric debate: the Verifier does not need RL training

A natural question is whether the debate gain requires both roles to be RL-trained, or whether a frozen base model in the Verifier role would suffice. Arm C tests this directly: solo-GRPO step-30 as Solver, frozen base Qwen3-4B as Verifier, 4-turn alternating debate.

Table 3: Arm C (asymmetric: solo-GRPO Solver + frozen base Verifier) on solo-GRPO step-30.

Cell	A single-shot	D self-cons k=4	E pass@4 oracle	C debate-asym	B debate-self
MATH-500 overall (n=500)	0.656	0.734	0.780	**0.814**	0.832
MATH-500 L4 (n=128)	0.617	0.695	0.773	0.812	0.820
MATH-500 L5 (n=134)	0.500	0.567	0.597	0.746	0.769
AIME 2024 (n=30)	0.133	0.100	0.167	0.200	0.233
AIME 2025 (n=30)	0.100	0.133	0.133	**0.433**	0.367

Bootstrap 95% CIs (paired, 2000 resamples) for the relevant comparisons on MATH-500:

Comparison	diff	95% CI	reads as
C vs D (matched-compute self-cons)	+8.0pp	[+5.0, +11.0]	clears 3pp gate cleanly
C vs pass@4 oracle	+3.4pp	[+0.4, +6.2]	clears 0; barely clears 3pp
C L5 vs D L5 (n=134)	+18.0pp	[+11.2, +24.6]	gain concentrates on hard
B vs C (RL Verifier vs base Verifier)	**+1.8pp**	**[−0.2, +3.8]**	**statistically null**
AIME 2025 B vs C (n=30)	−6.7pp	[−20.0, +6.7]	null (n too small)

Two things stand out. First, arm C with a frozen base Verifier still beats matched-compute self-consistency by 8.0 points on MATH-500 overall and by 18.0 points on L5, with non-overlapping CIs, and still beats the pass@4 oracle by 3.4 points (lower bound +0.4). Second, the RL-trained Verifier in arm B is not statistically better than the frozen base Verifier in arm C: B−C = +1.8 points with a CI of [−0.2, +3.8] that crosses zero. The point estimate favors arm B but the margin is small.

The active mechanism is the scaffold structure rather than the Verifier's RL training. The 4-turn alternation with role-conditioned system prompts does the work; RL-training the critic adds at best a small marginal lift. This fits the picture from Section 5.2: debate exceeds the pass@k oracle because critique-conditioning generates new trajectories rather than aggregating over old ones, and the critique need not come from a specialized policy to produce them. Practically, the scaffold is portable: the base model already contains the latent critique skill the Verifier role exercises, so a separately trained Verifier is not needed for deployment.

5.5 Self-consistency fails catastrophically on hard problems; debate does not

Cell	Single-shot	Self-cons k=4	Δ
Base AIME 2024 (n=30)	0.033 (1/30)	0.100 (3/30)	+6.7
Base AIME 2025 (n=30)	0.067 (2/30)	**0.000 (0/30)**	**−6.7**
Solo-GRPO AIME 2024 (n=30)	0.133 (4/30)	0.100 (3/30)	−3.3
Solo-GRPO AIME 2025 (n=30)	0.100 (3/30)	0.133 (4/30)	+3.3

Self-consistency on base Qwen3-4B AIME 2025 gets zero of thirty problems correct, worse than the two of thirty that single-shot manages. With $k = 4$ samples per problem, majority vote concentrates probability mass on the dominant wrong mode; on AIME 2024 it slightly hurts solo-GRPO too. The pattern matches the recent critique literature, but seeing it on AIME-class problems complicates the picture in which self-consistency is treated as a safe baseline.

Debate lifts the same cells across the board. On AIME 2024, base Qwen3-4B goes from 0.033 single-shot to 0.367 in debate; solo-GRPO goes from 0.133 to 0.233. On AIME 2025, base goes from 0.067 to 0.333; solo-GRPO goes from 0.100 to 0.367. The sequential conditioning structure is robust to the dominant-wrong-mode failure that breaks parallel self-consistency.

5.6 The gain is difficulty-conditioned

On solo-GRPO, the B−D margin is zero on L1 and L2 (where the Solver is already near ceiling), 5.7 points on L3, 12.5 points on L4, and 20.2 points on L5. The pattern is consistent with a ceiling-aware reading: where the Solver is nearly perfect, neither matched-compute baseline has anything to extract; where the Solver has headroom, debate extracts substantially more than self-consistency does.

Base Qwen3-4B has more headroom across the board (L5 single-shot at 0.299), and the per-level margins are larger: 0.0 on L1 (ceiling), +4.4 on L2, +16.2 on L3, +28.1 on L4, and +41.0 on L5 (Table 1, bootstrap CI [+32.1, +50.0]). The same monotone-in-difficulty pattern shows up on Qwen3-1.7B base (+2.3, +8.9, +16.2, +28.1, +32.8 on L1 through L5). Across all three checkpoints, debate's marginal value grows with problem difficulty until the model lacks enough knowledge to be critiqued at all.

5.7 Turn-count ablation: debate saturates at 4 turns

Why 4 turns? We ran the same scaffold at 2-turn (Solver T0, Verifier T1) and 6-turn (extending the alternation) on solo-GRPO MATH-500.

Table 4: Turn-count ablation on solo-GRPO MATH-500 ($n = 500$).

Variant	Overall	L1	L2	L3	L4	L5
Single-shot (A)	0.656	0.837	0.767	0.733	0.617	0.500
Self-cons k=4 (D)	0.734	0.907	0.856	0.819	0.695	0.567
Pass@4 oracle (E)	0.780	0.930	0.889	0.867	0.773	0.597
2-turn debate-self	0.804	0.907	0.822	0.857	0.820	0.702
4-turn debate-self	**0.832**	0.907	0.856	0.876	0.820	**0.769**
6-turn debate-self	0.832	0.930	0.867	0.867	0.836	0.746

The 2-turn protocol already captures 84% of the 4-turn lift over single-shot (14.8 of 17.6 points) at half the generation budget, and still exceeds the pass@4 oracle by 2.4 points overall and on every per-level cell below ceiling. A single Verifier turn is sufficient for most of debate's value, which matters for deployment. Going from 4 to 6 turns then produces no aggregate change (0.832 in both cases) and only noise-level per-level deltas (+2.3 on L1, +1.6 on L4, -2.3 on L5, well within sampling variance at $n = 43$ to 134 per level). The iterative correction is fully captured by the 4-turn protocol; additional turns do not help.

The 2-turn $\rightarrow$ 4-turn $\rightarrow$ 6-turn saturation curve (0.804 $\rightarrow$ 0.832 $\rightarrow$ 0.832) is what bounds the contribution claim cleanly. The 4-turn scaffold sits at the saturation point for this debate value at this model scale, and compute scaling beyond four turns runs into the iterative-correction loop's diminishing returns rather than additional accuracy.

5.8 Per-turn trajectory analysis: debate corrects 25-36% of problems with near-zero regression

We classify each problem's debate trajectory by initial-Solver (T0) correctness vs final (T3) correctness:

- **PRESERVED:** T0 correct $\rightarrow$ T3 correct.
- **CORRECTED:** T0 wrong $\rightarrow$ T3 correct (the headline mechanism).
- **REGRESSED:** T0 correct $\rightarrow$ T3 wrong (failure mode).
- **UNFIXED:** T0 wrong $\rightarrow$ T3 wrong.

Table 5: Trajectory categories on MATH-500 ($n = 500$ per row).

Checkpoint	PRESERVED	**CORRECTED**	REGRESSED	UNFIXED
Solo-GRPO step-30	57.8%	**25.4%**	**0.2%**	16.6%
Base Qwen3-4B	47.8%	**36.0%**	**0.2%**	16.0%
Base Qwen3-1.7B	46.6%	**31.0%**	**0.2%**	22.2%

The correction-to-regression ratio is sharply one-sided. Across all three checkpoints, debate flips 25 to 36 percent of wrong T0 answers to correct, while only 0.2 percent of correct T0 answers regress. The scaffold is almost never harmful and very often helpful, which fits the pass@k-exceeding result of Section 5.2 and rules out a noisy high-variance correction story.

What the verdict label is doing is less obvious. The Verifier's `<verify>approve|reject</verify>` tag is largely uncorrelated with Solver correctness. On solo-GRPO MATH-500 the Verifier approves a wrong Solver T0 30.8% of the time and only rejects a wrong T0 2.2% of the time; as a binary classifier the Verifier is poor. On Qwen3-1.7B base the Verifier omits the tag entirely on 81% of problems. Yet the scaffold still corrects a quarter to a third of problems on these same checkpoints. The mechanism cannot be the verdict tag or the binary critique label; it must be the Verifier's reasoning text, which the next Solver turn conditions on regardless of the explicit verdict. This is what Section 5.4 already pointed at: a frozen base Verifier that has never been RL-trained produces critique text indistinguishable in effect from an RL-trained Verifier's.

The implication for the multi-agent debate literature is that the "Verifier as a classifier" framing targets the wrong intermediate variable. Research effort spent improving Verifier verdict accuracy via auxiliary reward channels (R_verdict, Self-Verify Eq. 5, J1) is improving a quantity largely decoupled from the scaffold's corrective power. What matters is whether the Verifier turn produces reasoning text rich enough to re-condition the next Solver turn toward correctness. A frozen base model with the Verifier system prompt already does that.

6 Training-time multi-agent: a complementary negative result

We also ran the DTCA training recipe (Section 3.2) to controlled completion: 30 PPO updates with KL anchoring, no entropy collapse, no token-repetition pathology. The trained Solver, evaluated turn-0 single-shot, lands at:

- MATH-500 turn-0: **0.634** (vs solo-GRPO single-agent GRPO: 0.656)
- Per-level lifts vs base: similar to solo-GRPO (Table 6).

Table 6: DTCA vs solo-GRPO per-level lift over base on MATH-500 ($n = 43$-134 per cell).

Level	Base	Solo-GRPO Δ	DTCA Δ (blog 4 bootstrap)
L1	0.907	−0.070	−0.008
L2	0.722	+0.045	+0.030
L3	0.629	+0.104	+0.124
L4	0.445	+0.172	+0.203
L5	0.299	+0.201	+0.147

The two methods produce Solvers indistinguishable on single-shot accuracy: the elaborate role-asymmetric multi-channel reward design does not exceed simple single-agent GRPO at matched compute.

DTCA's full-debate (turn-3) accuracy of 0.574 is only marginally above the base single-shot accuracy of 0.534, materially below single-agent GRPO single-shot (0.656), and materially below inference-time debate on the untrained base model (0.838, Table 1). The training-time multi-agent system produces a 4-turn debate output that an inference-time 4-turn debate on the base model exceeds by 26.4 points. The Verifier itself has two characterized pathologies that explain the underperformance. On Verifier turns that emit a `<verify>` tag, 96.5% emit `approve`; the class-reweighted R_verdict reduces but does not eliminate this rubber-stamp behavior at the imbalanced Solver base rate (≈ 0.65 on MATH L3-5). And 53.4% of Verifier turns omit the tag entirely, which is the cheapest exploit against R_verdict: drop the tag, drop the verdict reward.

The raw-bypass per-turn format penalty (Section 3.2) is a transferable pattern for normalized multi-channel RL when channel variance can collapse to zero under format collapse and disable the in-channel penalty. We do not claim it solves the training-time multi-agent problem in general; the approval-bias pathology is more fundamental and remains. But the format-collapse fix is a publishable methodological primitive on its own. Taken together with single-agent GRPO matching DTCA on Solver single-shot, and with inference-time debate substantially beating matched-compute self-consistency, the picture is consistent: the productive multi-agent work for this benchmark happens at inference time rather than in the training objective.

7 Discussion

Why does the strict 4-turn role-asymmetric scaffold work where prior multi-agent debates have not? Three structural differences from typical literature setups are candidates. First, the protocol is strict and deterministic (Solver-Verifier-Solver-Verifier with four turns) rather than free-form. Second, role differentiation is supplied entirely through system prompts rather than through separate checkpoints or fine-tuned roles. Third, our baseline comparison is matched-token self-consistency at $k = 4$, not a weaker single-shot or unmatched baseline that some prior pro-multi-agent papers used. The base-model B$-$D margin of +22.4 points is hard to reconcile with any "longer thinking budget" story, because matched-token self-consistency already provides the longer thinking.

Why does base plus debate match solo-GRPO plus debate? Two complementary observations point at the answer. First, debate at four turns exceeds the pass@4 oracle by 5 to 30 points depending on cell. The Verifier turn is not selecting from k parallel samples; it is generating new trajectories by conditioning the next Solver turn on a critique. The base and RLVR-trained models have rather different parallel-sampling distributions (the base is higher-entropy), but both contain enough latent critique capacity that critique-conditioning extracts similar terminal accuracy. Second, the "internal pass@k with self-correction" mental model is therefore the wrong picture. The scaffold is closer to a one-step amortized search over conditioning sequences, with the Verifier emitting the conditioning string and the next Solver turn re-sampling under it. RLVR concentrates the prior, but the conditioning step is comparably effective in both regimes.

Why does self-consistency hurt on hard problems? When the model's modal samples are wrong, which happens once single-shot accuracy drops to around 0.3 on AIME-class problems, majority vote at $k = 4$ deterministically picks the modal wrong answer. The base AIME 2025 result is the canonical demonstration: single-shot reaches 2 of 30, self-consistency 0 of 30. Each of the two single-shot correct cases produced one correct and three wrong samples, and majority vote flipped them to wrong. Debate does not have this failure mode because it conditions each Solver turn on the Verifier's critique rather than on k parallel samples of the same distribution.

The DTCA result, taken together with the inference-time finding, sharpens the framing. With the strongest role-asymmetric multi-channel reward design we could put together, the Solver is indistinguishable from single-agent GRPO at matched compute. The multi-agent literature has often been asking whether multi-agent training helps. The more productive question, on this benchmark and family, is whether multi-agent inference time in a controlled scaffold helps over matched-compute single-agent. Our answer to the second is yes by 10 to 22 points on Qwen3-4B math reasoning.

Limitations and caveats are in Section 8.

8 Limitations

- **Family-level generalization (partial).** Primary results are on Qwen-family models (1.7B and 4B). The Qwen family is known to be unusually RLVR- and prompt-sensitive [13]. Cross-family probes on Phi-3.5-mini-instruct, SmolLM2-1.7B-Instruct, and Mistral-7B-Instruct-v0.3 are reported as supplementary in Appendix E; each hit a different infrastructure blocker and only directional evidence (B $-$ D = +8pp on Phi-3.5) is available. Generalization to Llama 3.x, Gemma 2, and other small models is open. The +21pp cross-scale-consistent B $-$ D result on Qwen-family bases is large enough that some effect should persist on other families, but quantitative claims will not transfer without testing.

- **Single domain (math).** AIME and MATH are math reasoning. GPQA-Diamond (science) is pending HuggingFace auth setup on Modal. Cross-domain transfer of the debate scaffold is open.

- **Single debate scaffold structure.** Strict Solver-Verifier alternation with role-prompt-only differentiation. We ablate turn count (2, 4, 6) on solo-GRPO MATH-500 (Section 5.7); role count (3+ agents), prompt-only vs fine-tuned roles, and non-alternating schedules are future work.

- **Asymmetric debate (arm C) on a single checkpoint.** Arm C ran successfully on solo-GRPO step-30 (Section 5.4) but was not run on Qwen3-1.7B or base Qwen3-4B due to budget discipline. The "Verifier does not need RL training" finding is therefore established on one checkpoint and inferred for others by the consistency of the trajectory analysis (Section 5.8).

- **AIME $n = 30$.** Per-year sample sizes are small. We report paired bootstrap CIs on the difference of proportions throughout, but per-cell bounds at $n = 30$ remain wide.

- **Bibtex placeholders.** All citation keys are placeholders. The specificity audit against IMAD, AgentArk, "Single-Agent Outperforms MAS Under Equal Thinking Token Budgets", and "Stop Overvaluing Multi-Agent Debate" (full results in `docs/specificity-check.md`) confirms no prior work tests our specific (Qwen-family, role-asymmetric, MATH-500/AIME, $k = 4$ matched-token self-consistency) slice, but final citation resolution is pending.

9 Conclusion

On Qwen3-4B math reasoning, a four-turn role-asymmetric debate scaffold at inference time substantially exceeds matched-compute self-consistency: +22pp aggregate on the base model, +10pp on a single-agent GRPO-trained checkpoint, +20-30pp on hard problems (L4-L5 MATH-500, AIME 2025). Both checkpoints converge to roughly the same debate accuracy, indicating RLVR is partially substitutable for the inference-time debate effect rather than synergistic with it. Self-consistency at matched compute catastrophically fails on uniformly-hard problems (base AIME 2025: 0/30), while debate maintains a positive lift. A complementary training-time experiment shows that an ambitious role-asymmetric multi-channel reward design (DTCA) produces a Solver indistinguishable from single-agent GRPO, so the productive multi-agent work is at inference, not at training.

A Implementation

5-arm matrix implementation: `dtca/eval/standalone.py`. Arm dispatch via `mode` field on `EvalConfig`; asymmetric arm loads two co-located vLLM engines at half `gpu_memory_utilization` each. Aggregation helpers `majority_vote` and `any_correct` unit-tested in `tests/test_eval/test_inference_arms.py` (23 tests, all passing).

CLI:

```
python scripts/evaluate.py \
    --checkpoint /path/to/checkpoint \
    --datasets math500 aime2024 aime2025 \
    --mode {single-shot|debate-self|debate-asymmetric|self-consistency|pass-at-k} \
    --k 4 \
    --verifier-checkpoint /path/base-model \
    --output results/exp_name/ \
    --limit 50    # cheap-probe slice
```

B Reproducibility

- Solo-GRPO training: Modal app `ap-WBGQTQ88X3EYIMDVkctxch`, experiment key `solo-grpo`, 30 PPO updates, ∼\$120 Modal compute.
- Eval runs: Modal `dtca-eval` app, multiple instantiations 2026-05-11. App URLs in commit log.
- Checkpoints: `/data/checkpoints/solo-grpo/global_step_30/actor/Qwen3-4B_SolverAgent` (FSDP-sharded), merged to `/data/merged/solo-grpo/global_step_30`. Base Qwen3-4B at `/data/models/Qwen3-4B`.
- Per-problem JSONL records under `/data/eval_results/{exp}/{step}/`, with mode-suffixed subdirectories (`mode-debate-self`, `mode-self-consistency-k4`, etc.) and per-dataset `transcripts.jsonl` files inside each.
- Commit hash: TBD on submission.

C Qualitative debate trajectories

The cleanest diagnostic of what debate adds beyond parallel-sampling aggregation is the set of problems on which the 4-turn debate produces the correct answer but all four independent samples in the pass@k arm are wrong. On solo-GRPO step-30 MATH-500 L5, 26 of 134 problems fall in this category. The supplementary file `docs/interesting-trajectories.md` walks through five examples; one representative case follows.

The problem (MATH-500 L5) asks: "The expression $2\cdot 3\cdot 4\cdot 5+1$ is equal to 121, since multiplication is carried out before addition. However, we can obtain values other than 121 by inserting parentheses (e.g. $(2\cdot(3\cdot 4))\cdot(5+1)=144$). In total, how many values can be obtained?" Ground truth: 4.

The four parallel pass@4 samples gave boxed answers of 60, 122, 126, and 144; none correct. Each of the four found a different way to misread the problem, counting expression evaluations rather than distinct numerical values. The debate trajectory unfolded differently. At turn 0 the Solver boxed 14, having reasoned through some parenthesizations but lost track of the "how many distinct values" framing. At turn 1 the Verifier critiqued by enumerating specific parenthesizations such as $(2 \cdot 3 \cdot 4) \cdot (5 + 1) = 144$ and $(2 \cdot (3 \cdot 4 \cdot 5 + 1)) = 122$; its own boxed answer of 126 was wrong, but its critique named the operation as enumerating groupings and reading off distinct numerical values. At turn 2 the Solver re-attempted with the Verifier's framing, listed ten parenthesizations, computed each, and collected the distinct values $\{121, 122, 126, 144\}$, boxing the correct answer 4. At turn 3 the Verifier approved with 4.

The Verifier's turn-1 critique did not contain the correct answer. What it contained was a framing of the task (enumerate groupings, count distinct values) that was the missing piece in the Solver's turn-0 reasoning. The Solver's turn 2 produced a correct answer that none of the four independent samples reached. This is the mechanism the per-cell pass@k-exceeding results in Section 5.2 are made of: the Verifier turn supplies reasoning structure the Solver did not surface single-shot, and the next Solver turn re-conditions on that structure. We expect this is the source of the roughly 30-point pass@k surplus on L5 problems specifically, where the model has the relevant knowledge but fails to surface it in any single sample at temperature 0.7. The full set of 26 cases is in `docs/interesting-trajectories.md`.

D DTCA failure-mode characterization

The two principal DTCA Verifier pathologies are characterized quantitatively in Section 6: a 96.5% rubber-stamp approval rate on the Verifier turns that emit a `<verify>` tag, and a 53.4% missing-tag rate on Verifier turns overall. The raw-bypass per-turn format penalty (Section 3.2) addresses the second pathology by adding a fixed -1.0 reward to per-turn advantages that bypasses GDPO per-channel normalization, restoring gradient pressure on format compliance that the in-channel penalty loses once the channel's empirical variance collapses to zero under format collapse. The approval-bias pathology is more fundamental: with Solver correctness at base rate ≈ 0.65 on MATH L3-5, always-approve has unconditioned expected reward ≈ 0.30 before class reweighting and remains positive after the EMA-based Self-Verify weights settle. We leave a principled fix to future work. The pre-RL Solver-T0 single-shot lift the DTCA recipe produces (Table 6) is therefore not attributable to a working Verifier; the Verifier's training signal is largely a slow constant that the GRPO update absorbs through the shared-policy Solver gradient, not via the role-asymmetric critique we designed.

E Cross-family probes (preliminary)

We attempted the 3-arm matrix on three non-Qwen models to test cross-family generalization of the inference-debate finding. Each model hit a different blocker; we report only the cells that ran cleanly.

microsoft/Phi-3.5-mini-instruct (3.8B, MIT). All three arms completed. Single-shot is uniformly weak (overall 0.050, L1 only 0.139, far below Phi-3.5's documented capability), indicating an answer-format/`\boxed{}` extraction mismatch on top of any capability ceiling. Self-consistency k=4 = 0.072 (modest lift). **Debate-self = 0.152 overall** (per-level L1=0.442, L2=0.367, L3=0.095, L4=0.094, L5=0.015), an $3\times$ lift over single-shot and $2\times$ over self-consistency. The L1 jump from 0.140 to 0.442 strongly suggests Phi-3.5's reasoning capacity is intact and debate is unlocking what single-shot extraction failed to surface. B$-$D margin = +8.0pp on MATH-500 overall, directionally consistent with our Qwen findings, though at lower absolute accuracy and confounded by the extraction-recovery effect. We do not include Phi-3.5 in the headline claims of the paper because the extraction confound prevents a clean cross-family attribution; the directional signal is reported here only.

HuggingFaceTB/SmolLM2-1.7B-Instruct (1.7B, Apache). Arm A single-shot landed at 0.078 overall on MATH-500 (per-level L1=0.256, L2=0.156, L3=0.076, L4=0.031, L5=0.015). Arm D self-consistency $k = 4$ landed at 0.106 (L1=0.279, L2=0.200, L3=0.114, L4=0.047, L5=0.037). Self-consistency provides a +2.8pp lift over single-shot overall. Arm B (4-turn debate-self) is structurally incompatible with SmolLM2's 8192-token context: the 4-turn debate accumulates $\sim$9860 tokens by turn 3 at `max_tokens=2048` per turn, exceeding the model's context window. A fair cross-family test on SmolLM2 would require reducing `max_tokens` per turn and re-running both arms B and D at matched lower budget. We do not report a SmolLM2 B $-$ D number.

mistralai/Mistral-7B-Instruct-v0.3 (7B, Apache). All three arms hit a vLLM 0.8.4 engine init bug (`TypeError: unsupported operand type(s) for *: 'int' and 'NoneType'`), likely related to the Mistral tokenizer-mode handling vLLM warns about. Future work: pass `tokenizer_mode='mistral'` and rerun.

In the current draft the cross-family probe is therefore incomplete. The main empirical claim is scoped to the Qwen family at 1.7B and 4B, with the documented expectation that some effect should persist on other families given the cross-scale-consistent B−D margin on the Qwen bases (+21 to +22 points). A revision once the cross-family infrastructure is fixed (per-model answer-parser variants, tokenizer-mode handling, matched-compute budget renormalization for short-context models) will report clean cross-family numbers.

F Full per-level results

Table 1 in the main text reports headline cells (MATH-500 overall, L4, L5, AIME 2024, AIME 2025). The full per-level accuracy table and per-cell bootstrap CIs are reproduced here.

Table F1. Per-cell accuracy on MATH-500 (per-level $n \in \{43, 90, 105, 128, 134\}$ for L1-L5) and AIME ($n = 30$ per year). A is single-shot, D is $k = 4$ self-consistency, B is 4-turn debate-self.

	A	D	B	B−A	B−D
Base Qwen3-4B					
Overall (n=500)	0.534	0.614	**0.838**	+30.4	+22.4
L1	0.907	0.907	0.907	+0.0	+0.0
L2	0.722	0.833	0.878	+15.6	+4.4
L3	0.629	0.705	0.867	+23.8	+16.2
L4	0.445	0.555	**0.836**	+39.1	+28.1
L5	0.299	0.358	**0.769**	+47.0	+41.1
AIME 2024	0.033	0.100	**0.367**	+33.3	+26.7
AIME 2025	0.067	0.000	**0.333**	+26.7	+33.3
Solo-GRPO step-30					
Overall (n=500)	0.656	0.734	**0.832**	+17.6	+9.8
L1	0.837	0.907	0.907	+7.0	+0.0
L2	0.767	0.856	0.856	+8.9	+0.0
L3	0.733	0.819	0.876	+14.3	+5.7
L4	0.617	0.695	**0.820**	+20.3	+12.5
L5	0.500	0.567	**0.769**	+26.9	+20.2
AIME 2024	0.133	0.100	**0.233**	+10.0	+13.3
AIME 2025	0.100	0.133	**0.367**	+26.7	+23.4
Base Qwen3-1.7B					
Overall (n=500)	0.520	0.564	**0.776**	+25.6	+21.2
L1	0.907	0.907	0.930	+2.3	+2.3
L2	0.678	0.767	0.856	+17.8	+8.9
L3	0.610	0.686	0.848	+23.8	+16.2
L4	0.477	0.484	**0.766**	+28.9	+28.1
L5	0.261	0.299	**0.627**	+36.6	+32.8
AIME 2024	0.000	0.033	**0.133**	+13.3	+10.0
AIME 2025	0.000	0.067	**0.267**	+26.7	+20.0

Table F2. Per-cell 95% paired bootstrap CIs on $p_B - p_D$ (2000 resamples each). Gate threshold is +3pp lower bound.

Table F3. Per-cell 95% paired bootstrap CIs on $p_B - p_{\text{pass@4}}$ (2000 resamples each). Hardest cells per checkpoint.

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Cell	$p_B - p_D$	95% CI
Base Qwen3-4B		
MATH-500 overall	+22.4	[+18.6, +26.2]
L1	0.0	[−7.0, +7.0]
L2	+4.4	[−1.1, +11.1]
L3	+16.2	[+8.6, +23.8]
L4	+28.1	[+20.3, +35.9]
L5	+41.0	[+32.1, +50.0]
AIME 2024	+26.7	[+13.3, +43.3]
AIME 2025	+33.3	[+16.7, +50.0]
Solo-GRPO step-30		
MATH-500 overall	+9.8	[+6.8, +13.0]
L1	0.0	[0.0, 0.0]
L2	0.0	[−5.6, +5.6]
L3	+5.7	[+1.9, +10.5]
L4	+12.5	[+5.5, +20.3]
L5	+20.2	[+13.4, +27.6]
AIME 2024	+13.3	[+3.3, +26.7]
AIME 2025	+23.3	[+10.0, +40.0]
Base Qwen3-1.7B		
MATH-500 overall	+21.2	[+17.4, +25.2]
L1	+2.3	[0.0, +7.0]
L2	+8.9	[+2.2, +15.6]
L3	+16.2	[+8.6, +23.8]
L4	+28.1	[+19.5, +35.9]
L5	+32.8	[+23.9, +41.8]
AIME 2024	+10.0	[0.0, +23.3]
AIME 2025	+20.0	[+6.7, +36.7]

Cell	$p_B - p_{\text{pass@4}}$	95% CI
Base Qwen3-4B MATH-500 overall	+14.6	[+11.2, +18.2]
Base Qwen3-4B MATH-500 L4	+18.8	[+10.9, +26.6]
Base Qwen3-4B MATH-500 L5	+29.9	[+21.6, +38.1]
Base Qwen3-4B AIME 2024	+23.3	[+10.0, +40.0]
Base Qwen3-4B AIME 2025	+20.0	[+6.7, +36.7]
Solo-GRPO MATH-500 overall	+5.2	[+2.2, +8.2]
Solo-GRPO MATH-500 L5	+17.2	[+9.7, +24.6]
Solo-GRPO AIME 2025	+23.3	[+10.0, +40.0]
Qwen3-1.7B MATH-500 overall	+10.2	[+7.0, +13.4]
Qwen3-1.7B MATH-500 L5	+19.4	[+11.2, +26.9]
Qwen3-1.7B AIME 2025	+16.7	[+3.3, +30.0]

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